

25 Java OOP + Collections Interview Questions for QA/SDET

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This guide covers essential Java concepts and collections-related questions for QA engineers and SDETs preparing for automation interviews.

1. What are the four pillars of OOP?

Answer:

Encapsulation, Inheritance, Polymorphism, and Abstraction.

2. What is encapsulation?

Answer:

Encapsulation hides internal data and exposes functionality through public methods. It is achieved by using private variables with public getters and setters.

3. What is inheritance?

Answer:

Inheritance allows one class to acquire properties and behaviors of another class using the extends keyword.

4. What is polymorphism?

Answer:

Polymorphism means the ability of an object to take many forms. It includes method overloading (compile-time) and method overriding (run-time).

5. What is abstraction?

Answer:

Abstraction hides implementation details and exposes only relevant functionality. Achieved using abstract classes and interfaces.

6. What is the difference between abstract class and interface?

Answer:

An abstract class can have both abstract and concrete methods, while an interface can only have abstract methods (prior to Java 8). Interfaces support multiple inheritance.

7. What is method overloading?

Answer:

Defining multiple methods with the same name but different parameter lists in the same class.

8. What is method overriding?

Answer:

Providing a new implementation for a method in a subclass that already exists in the parent class.

9. What is the difference between final, finally, and finalize()?

Answer:

- final: keyword for constants, methods, and classes
- finally: block that always executes after try-catch

- `finalize()`: method called by garbage collector before object destruction
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10. What is a constructor?

Answer:

A constructor is a special method used to initialize objects. It has the same name as the class and no return type.

11. What is a collection in Java?

Answer:

A collection is a framework that provides classes and interfaces for storing and manipulating groups of data as a single unit.

12. What are the main types of collections?

Answer:

List, Set, Queue, and Map.

13. What is the difference between ArrayList and LinkedList?

Answer:

ArrayList uses dynamic arrays and is faster for random access. LinkedList uses a doubly linked list and is faster for insertions and deletions.

14. What is the difference between HashSet and TreeSet?

Answer:

HashSet does not maintain order and is faster. TreeSet maintains elements in sorted order but is slower.

15. What is a HashMap?

Answer:

HashMap stores key-value pairs with unique keys. It allows null values and one null key.

16. How does HashMap work internally?

Answer:

HashMap uses a hash function to compute the index and stores entries in buckets. Handles collisions using linked lists or trees (since Java 8).

17. What is the difference between HashMap and Hashtable?

Answer:

HashMap is not synchronized and allows one null key. Hashtable is synchronized and does not allow null keys or values.

18. What is the difference between List and Set?

Answer:

List allows duplicates and maintains insertion order. Set does not allow duplicates and may or may not maintain order.

19. What is the difference between Iterator and ListIterator?

Answer:

Iterator can traverse in one direction and remove elements. ListIterator can traverse in both directions and add elements.

20. What is the difference between Comparable and Comparator?

Answer:

Comparable provides natural ordering in a class by implementing `compareTo()`. Comparator provides custom ordering by implementing `compare()`.

21. What is a synchronized collection?

Answer:

A synchronized collection is thread-safe and prevents concurrent modification by multiple threads.

22. What is a fail-fast iterator?

Answer:

An iterator that throws `ConcurrentModificationException` if the collection is modified while iterating.

23. How do you convert an array to a list in Java?

Answer:

Use `Arrays.asList(array)`.

24. How do you sort a collection?

Answer:

Use `Collections.sort(list)` for natural order or `Collections.sort(list, comparator)` for custom order.

25. Why is understanding Java OOP and collections important for QA?

Answer:

Automation frameworks often use Java. Understanding OOP and collections helps QA engineers write efficient, reusable, and maintainable automation scripts.